Samantha Rodriguez, CFA, is employed at a firm that specializes in assisting institutional investors, including pension funds, in hiring the most qualified portfolio managers for their investment mandates. Because of her heavy workload, she is also hiring an associate to help her. One of the candidates she interviewed was Karla Jang, who made the following statements during the interview:

Statement 1:	Type II errors in hiring and firing investment managers are easier to identify than Type I errors.
Statement 2:	In an efficient market, the costs of hiring or retaining weak managers are higher than the costs in a less efficient market.
Statement 3:	Given that Type I errors result in explicit costs, clients are more likely to be bothered by Type I errors than by Type II errors.

Rodriguez ultimately decides to hire another candidate, Frank Mompou, as her associate; his first assignment is to help an institutional investor with a long time horizon and high risk tolerance in hiring an investment manager to maximize returns.

Rodriguez is helping a pension fund manager, Brenna Quan, understand the way performance fee structures affect the incentives of her investment managers and the volatility of net returns. During their discussions, Quan makes the following statements:

Statement 4:	Performance-based fees benefit investors because they will pay relatively lower fees when there are lower active returns.
Statement 5:	Symmetrical fee structures are preferred by managers over performance-based fees.
Statement 6:	A performance-based fee with a base fee and a maximum fee will incentivize risk taking, as it is analogous to the investment manager being long a call option.

One of Quan's investment managers has the following fee structure:

Table 1: Performance-Based Fee Structure	
Standard fee	0.60%
Base fee	0.30%
Sharing	20.00%

There is both a minimum and a maximum annual billed fee in place. The annual billed fee is calculated partly on the active return in excess of the base fee percentage.

Question #1 - 4 of 45

Question ID: 1675002

How many of the statements made by Jang during the job interview are correct?

A) None.

B) One.

C) Two.

Question #2 - 4 of 45

Question ID: 1675003

How should Mompou evaluate drawdowns and capture ratios in hiring an investment manager for the institutional investor, taking into account both positive and negative returns of the relevant benchmark?

A) Mompou should hire managers with a history of large drawdowns and drawdown durations.

B) Mompou should hire managers with a history of large upside and downside capture ratios.

C) Mompou should hire managers with a history of large drawdowns and drawdown durations, as well as large upside and downside capture ratios.

Question #3 - 4 of 45

Question ID: 1675004

How many of the statements made by Quan during the discussion are correct?

A) One.

B) Three.

C) Two.

Question #4 - 4 of 45

Question ID: 1675005

Based on Table 1, what is the required breakeven active return and the resulting maximum annual billed fee when there is symmetry within the active return range centered around the breakeven active return?

	Breakeven active return	Maximum annual billed fee
A)	1.80%	0.60%
B)	1.80%	0.90%
C)	1.50%	0.60%

Question #5 of 45

Question ID: 1674962

When discussing Type I and Type II errors, an analyst notes that errors of omission and opportunity costs can be associated with these errors. These considerations *best* describe:

	Errors of omission	Opportunity costs
A)	Type I error	Type I error
B)	Type II error	Type II error
C)	Type II error	Type I error

Question #6 of 45

Question ID: 1674961

An example of a Type I error is:

A) not firing a manager who outperforms expectations.

B) retaining a manager who underperforms expectations.

C) firing a portfolio manager who outperforms expectations.

Question #7 of 45

Question ID: 1674994

A performance-based fee structure where the manager's fee consists of a base fee and sharing positive portfolio performance is analogous to a manager having a long call option:

A) on the portfolio's return with a strike price equal to the base fee.

B) and a long put option on the portfolio's return both with a strike price equal to the base fee.

C) on the portfolio's return with a strike price equal to the base fee, and a short call option on the portfolio's return with a strike price equal to the maximum fee.

Question #8 of 45

Question ID: 1674976

The ABC Fund uses equity-based strategies but has a very high turnover. When considering the use of either a returns-based style analysis (RBSA) or a holdings-based style analysis (HBSA), the *most appropriate* style analysis for each of the factors is:

	Equity-based strategies	High turnover
A)	RBSA	HBSA
B)	HBSA	RBSA
C)	HBSA	HBSA

Question #9 of 45

Question ID: 1674982

Last month, a fund manager's upside and downside captures were both 80%. This implies that the manager underperformed the benchmark:

A) only when the benchmark return was negative.

B) regardless of market movement.

C) only when the benchmark return was positive.

Question #10 of 45

Question ID: 1674975

Considering the investment manager selection process, holdings-based style analysis (HBSA) is *most appropriate* to use when evaluating:

A) portfolios with high turnover.

B) illiquid securities.

C) equity portfolios.

Question #11 of 45

Question ID: 1674990

One advantage of separately managed accounts (SMAs) over pooled investments is:

A) greater tax efficiency.

B) lower trading costs.

C) less transparency in demonstrating value added.

Question #12 of 45

Question ID: 1674963

Which of the following would be regarded as the *least appropriate* method to measure the performance of a hedge fund?

A) The Sharpe ratio.

B) Separate long/short benchmarks.

C) Relative performance comparisons with traditional benchmarks.

Question #13 of 45

Question ID: 1674965

Jack Jensen is the president of Jensen Management. Jensen prides himself on the care of his employees. He states that in 30 years of portfolio management, he has only had to fire two employees. Tom Mercer is president of Analytical Investors. His policy has been to replace poorly performing managers, where poor performance equals underperforming their benchmark for two successive quarters. Which of the following *best* describes these managers' continuation decisions?

A) Jensen is not likely to be committing any error, and Mercer is likely committing Type II error.

B) Jensen is likely committing Type I error, and Mercer is likely committing Type II error.

C) Jensen is likely committing Type II error, and Mercer is likely committing Type I error.

Question #14 of 45

Question ID: 1674980

An analyst observes that a portfolio's capture ratio (CR) is 0.65. This would *most likely* indicate:

A) negative return asymmetry.

B) a convex return profile.

C) positive return asymmetry.

Question #15 of 45

Question ID: 1674960

When considering a manager's philosophy, we are *most likely* to be concerned with that manager's:

A) strategy feasibility in current market conditions.

B) likelihood of strategy execution, given the manager’s expertise.

C) targeted market inefficiency for earning excess returns.

Question #16 of 45

Question ID: 1674968

A mutual fund CEO is reviewing the performance of her fund managers. The CEO assumes that markets are mean reverting. Which of the following statements about Type I or Type II errors is *least accurate* under mean-reverting markets?

A) Firing a strongly performing manager too soon is an example of a Type I error.

B) Firing a weakly performing manager too soon is an example of a Type II error.

C) Hiring or retaining a manager who subsequently underperforms is an example of a Type I error.

Question #17 of 45

Question ID: 1674966

John Peterson is trying to decide whether to retain his investment manager. His null hypothesis is that his current manager is not skillful. Peterson would be correct in stating that in relation to Type I errors, Type II errors are *most likely* to:

A) lead to explicit costs.

B) be apparent to investors.

C) be errors of omission.

Question #18 of 45

Question ID: 1675000

The fee schedule of an investment fund includes a base fee plus sharing both positive and negative fund performance. The sharing is based on returns net of the base fee. The investment fund has a(n):

A) fee structure similar to a long call option on the fund's return with a strike price equal to the base fee.

B) asymmetrical fee structure because of the base fee.

C) symmetrical fee structure.

Question #19 of 45

Question ID: 1674977

Due to a recent shift in asset allocation, the Grandvale Fund has a significant exposure to less liquid real estate and private equity investments. Because the portfolio contains illiquid securities, relative to the holdings-based style analysis (HBSA), the returns-based style analysis (RBSA) is:

A) neither more nor less accurate.

B) more accurate.

C) less accurate.

Question #20 of 45

Question ID: 1674978

In discussing style analysis, a portfolio manager makes the following two statements:

Statement 1:	"A holdings-based style analysis allows me to estimate a portfolio's sensitivities to security market indices."
Statement 2:	"I prefer the holdings-based style analysis because of its lower computational requirements and less complexity."

The manager is correct with respect to:

A) Statement 2 only.

B) Statement 1 only.

C) neither Statement 1 nor 2.

Question #21 of 45

Question ID: 1674967

The wider the dispersion of returns between a weak and a strong fund manager, the *less likely* there will be:

A) both Type I and Type II errors.

B) Type I error only.

C) Type II error only.

Question #22 of 45

Question ID: 1674983

The three-year historical performance indicators of a fixed-income portfolio produce an upside capture of 1.15 and a downside capture of 0.85. The capture ratios indicate that the portfolio:

A) has negative asymmetry.

B) has a concave return profile.

C) will outperform the benchmark in any market condition.

Question #23 of 45

Question ID: 1674981

John Addams manages three large-cap equity funds. Fund A has a capture ratio (CR) of 1.6, Fund B's is 1.0, and Fund C's is 0.8. Which fund has a concave return profile?

A) Fund B.

B) Fund A.

C) Fund C.

Question #24 of 45

Question ID: 1674974

Considering the investment manager selection process, style analysis is *most appropriate* when examining a manager who invests in:

A) publicly traded stocks.

B) complex derivatives.

C) alternative investments such as private equity.

Question #25 of 45

Question ID: 1674959

Which of the following approaches is *most appropriate* to use in the process of developing a manager universe?

A) Capture ratio analysis.

B) Performance assessment.

C) Holdings-based style analysis.

Question #26 of 45

Question ID: 1674988

Relative to pooled investments, a disadvantage of separately managed accounts (SMAs) is that SMAs:

A) have less ability for customization to client-specific objectives.

B) have higher transaction costs.

C) lower tax efficiency.

Question #27 of 45

Question ID: 1674993

An investor wants to select a performance-based fee structure that provides the greatest alignment between the investor's and manager's incentives. The fee structure that *best* meets this requirement is a:

A) bonus with full upside but limited downside exposure.

B) bonus with limited upside and downside exposures.

C) symmetrical structure with full upside and downside exposures.

Question #28 of 45

Question ID: 1674987

A securities analyst makes the following two statements about pooled and separately managed accounts (SMAs):

Statement 1:	Although SMAs allow investors better customization than pooled accounts, SMAs lack transparency and position-level details.
Statement 2:	Because SMAs are managed by dedicated managers, they are less subject to micromanagement risk by investors.

How many of the manager's statements are correct?

A) None.

B) Two.

C) One.

Question #29 of 45

Question ID: 1674964

Suppose that a portfolio management firm has decided that the costs of firing managers are excessive. Which of the following would be their *most appropriate* course of action? The firm should:

A) tolerate more Type II error to reduce Type I error.

B) tolerate more Type I error to reduce Type II error.

C) reduce both Type I and Type II errors.

Miguel Bahiense is the marketing director of Porrada Capital Management (PCM). After conducting recent research of its clients, competitors, and markets, Bahiense concludes that investors are increasingly sensitive to management fees. Bahiense and his team have developed a new, optional performance-based fee structure as an alternative to the PCM standard fees.

Bahiense prepares a presentation to the board to illustrate the proposed performance-based fee structure and compiles the following details, shown in Exhibit 1.

Exhibit 1: Fee Structure

Fee Details	PCM Standard Fee	PCM Performance-Based Fee
Fixed % of NAV	0.60%	–
Base fee	–	0.30%
Minimum fee	–	0.30%
Performance fee on active return beyond the base fee	–	20%
Maximum total fee	–	0.85%

Helena Azevedo, a portfolio analyst at PCM, compiles performance data to illustrate a comparison between the traditional PCM standard fees and the proposed new, optional performance-based fees. Over the 12-month measurement period, the PCM Portfolio X achieved a 12.5% return, against a benchmark of 9.2%.

At the board meeting, Ferenanda Lima, Head of Risk, asks, "At what rate of active return is the breakeven between the two fee structures?" Azevedo notes this request for additional information from Lima.

Board member Larissa Costa, Head of Quantitative Strategies, likes the proposal and explains that this performance-based fee structure can be replicated using options.

Question #30 - 33 of 45

Question ID: 1674996

Calculate the total billed fee for Portfolio X, assuming the client has elected the performance-based fees.

A) 0.60%.

B) 0.85%.

C) 0.90%.

Question #31 - 33 of 45

Question ID: 1674997

Calculate the level of active return that results in breakeven between the standard and the performance-based fee structures.

A) 1.80%.

B) 1.60%.

C) 1.70%.

Question #32 - 33 of 45

Question ID: 1674998

If the fee structure was replicated using options, which of the following is *closest* to the payoff diagram created?

A) Long call on the standard fee.

B) Long call on the base fee.

C) Bull call spread.

Question #33 - 33 of 45

Question ID: 1674999

Which of the following return combinations results in a total fee of 0.74% using the performance-based fee method?

A) Benchmark return of –5.6%; fund return of –2.90%.

B) Benchmark return of 6.40%; fund return of 8.90%.

C) Benchmark return of 8.80%; fund return of 10.80%.

Question #34 of 45

Question ID: 1674986

Behavioral inefficiencies are *most likely* to be:

A) temporary.

B) most common when regulations are changing.

C) long lived in nature.

Lucy Wong is head of manager selection at FundNet, a successful online fund-of-funds investment platform. FundNet applies a rigorous manager selection process to identify, analyze, select, and monitor investment management firms. FundNet aims to identify some of the best managers in each investment universe over a market cycle, typically three to five years.

FundNet segments investment firms into three categories: traditional long-only funds, liquid alternatives, and private equity funds. Keiko Wu, a due diligence analyst at FundNet, has been asked to review the manager selection performance data over the last 12 months for the traditional long-only category. Wu collects the relevant data and incorporates hypothesis testing to review the significance of the skill of each manager, the null hypothesis being a manager has no skill. Wu also reviews the firm's policy on Type I and Type II errors in her analysis. Wu writes up her report and makes the following observations.

Observation 1:	Our current significance level of our hypothesis test is 5%. I recommend that we change this to 2.5%.
Observation 2:	I have identified that we have missed out on a few excellent managers over the period, who we failed to identify as being skillful as part of our process.
Observation 3:	Our management selection process may include behavioral biases —specifically, feelings of regret in bad decision-making.
Observation 4:	The long-only funds, in theory, should show lower costs of Type I and Type II errors compared to the liquid alternatives and private equity markets.
Observation 5:	Type I and II errors both create explicit costs for the firm.

Wong reviews Wu's report, and then arranges a meeting to explore Wu's observations in further detail.

Question #35 - 38 of 45

Question ID: 1674970

Wong asks Wu to explain the impact of her recommendation in Observation 1 on Type I and Type II errors.

A) Type I errors will decrease, and Type II errors will increase.

B) Type I errors will increase, and Type II errors will decrease.

C) Type I errors and Type II errors will decrease.

Question #36 - 38 of 45

Question ID: 1674971

In relation to Observation 2, which of the following would assist in identifying this issue?

A) Type I errors.

B) Type II errors.

C) Neither Type I nor Type II errors.

Question #37 - 38 of 45

Question ID: 1674972

In relation to Observation 3 and behavioral theory, which of the following is *most accurate*?

A) Type II errors are psychologically more painful than Type I errors.

B) Type I errors are psychologically more painful than Type II errors.

C) Both errors are equally psychologically painful.

Question #38 - 38 of 45

Question ID: 1674973

Which of the following is *most accurate* of Observations 4 and 5?

A) Neither observation is accurate.

B) Only Observation 4 is accurate.

C) Only Observation 5 is accurate.

Question #39 of 45

Question ID: 1674979

An investment manager uses capture ratios to determine how suitable a fund is to an investor's risk tolerance and time horizon. The manager notes that the investment strategy is low beta, which would *most likely* indicate that the strategy will have a:

A) higher up capture and a higher down capture.

B) lower up capture and a lower down capture.

C) lower up capture and a higher down capture.

Question #40 of 45

Question ID: 1674985

The fund manager of Betafund's equity portfolio uses hard stop-loss orders to protect against adverse price changes. The primary disadvantage of this hedging strategy is that:

A) it involves subjective evaluation.

B) positions may be closed too frequently.

C) it does not protect against large losses.

Question #41 of 45

Question ID: 1674984

In equity markets where return volatility for individual securities is increasing relative to the volatility of the market portfolio, value-driven strategies are *most likely* to have:

A) lower upside capture than momentum strategies do.

B) higher upside capture than momentum strategies do.

C) similar upside and downside capture to that of momentum strategies.

Question #42 of 45

Question ID: 1674992

The payoff to investment managers using a performance-based fee schedule with upside limits is similar to the payoff diagram of a:

A) bear call spread.

B) short collar.

C) bull call spread.

Question #43 of 45

Question ID: 1674989

Which of the following statements regarding differences in separately managed accounts (SMAs) and pooled investments is accurate?

A) A customized SMA can help minimize tracking error relative to a benchmark.

B) The incremental cost of adding additional investors is relatively low for pooled investments.

C) Pooled investments have higher transaction costs compared to SMAs but can benefit from customization.

Question #44 of 45

Question ID: 1674958

Assigning a manager to a benchmark using objective risk exposures that are derived from the manager's actual return series is *best* captured under the:

A) holdings-based style analysis.

B) returns-based style analysis.

C) manager experience.

Question #45 of 45

Question ID: 1674991

An investment in limited partnerships is *most appropriate* for an investor with a:

A) high liquidity need.

B) long time horizon.

C) desire to benefit from tactical asset allocation.